DO NOW — RECALL

Lesson 077 · Linear Search: Code and Trace

Name: _____ Class: _____ Date: _____

1. Can linear search work with unsorted input?

2. What is not-found sentinel?

3. What does O(n) time mean?

4. Array a=[6,2,9,4]: set a[2]=a[0]+a[1]. Give updated array and total.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Yes.

2. Model answer

Distinct return such as −1.

3. Model answer

Work grows linearly with n.

4. Model answer

[6,2,8,4], sum20.

By the end, you can…

• Visit each array value in order; linear search does not require sorted data.

• Return the first matching index or a distinct not-found value such as −1.

• Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

Vocabulary — know these terms

linear search — Sequentially compare elements.

sentinel — Special result indicating a condition.

index — Zero-based position.

NEW LEARNING

Read, annotate and explain

Understand

Visit each array value in order; linear search does not require sorted data.

Trace

Return the first matching index or a distinct not-found value such as −1.

Explain

Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find 6 in unsorted [1,5,6,3].

Check the working

Check indexes0,1,2, return2; stop before3.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace 2 in [4,9,2,7], give checks and return. [6]

Question 2

Write C# linear search returning index/−1. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

State linear search best/worst time, space and one edge test. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Linear search requires a pre-sorted array.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.